WEEK 4 MACHINE LEARNING PROJECT REPORT

Advanced Model Tuning and Feature Engineering

Project: Student Placement Prediction

B.Tech – Computer Science Engineering

Machine Learning Project

This report documents the feature engineering, hyperparameter tuning, multi-algorithm experimentation, performance comparison, and final model selection performed during Week 4.

1. Executive Summary

The objective of Week 4 was to improve and critically evaluate the Student Placement Prediction model developed during Week 3. The original Logistic Regression model was revisited as the baseline. Feature scaling and categorical encoding were retained, while interaction features were introduced to capture potential relationships between student attributes. Hyperparameter tuning was then performed using GridSearchCV with 5-fold cross-validation. In addition, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) were evaluated in both default and tuned configurations.

The experiments showed that the Week 3 Logistic Regression baseline remained the strongest model on the held-out test set, achieving 80.85% accuracy and a 77.27% F1 score. Feature engineering and Logistic Regression tuning did not improve the baseline F1 score. Random Forest improved after tuning but remained below the baseline, while SVM produced the second-best overall F1 score in its default configuration. KNN improved substantially after tuning but remained the weakest family of models overall.

2. Objective

The main objectives of this phase were to:

• Revisit and establish the Week 3 Logistic Regression baseline.

• Apply feature scaling, categorical encoding, and interaction-based feature engineering.

• Perform systematic hyperparameter tuning using GridSearchCV and 5-fold cross-validation.

• Compare multiple classification algorithms using accuracy, precision, recall, and F1 score.

• Identify the most appropriate model for Student Placement Prediction based on experimental evidence.

3. Dataset Description

The cleaned dataset contains 10,000 student records and 12 columns. The target variable is PlacementStatus, with the classes Placed and NotPlaced. StudentID was excluded from model training because it is an identifier rather than a meaningful predictive feature.

Features used: CGPA, Internships, Projects, Workshops/Certifications, AptitudeTestScore, SoftSkillsRating, ExtracurricularActivities, PlacementTraining, SSC_Marks, and HSC_Marks.

The target was encoded as 1 for Placed and 0 for NotPlaced. Numerical variables were standardized, while categorical variables were transformed using one-hot encoding.

4. Methodology

4.1 Data Preparation

The dataset was loaded into a Pandas DataFrame. StudentID and PlacementStatus were separated from the predictive feature matrix. The data was split into 80% training and 20% testing subsets using stratification and random_state=42.

4.2 Preprocessing

Numerical variables were processed using StandardScaler. Categorical variables were processed using OneHotEncoder(handle_unknown='ignore'). A Scikit-learn Pipeline and ColumnTransformer were used to keep preprocessing consistent and avoid data leakage during cross-validation.

4.3 Feature Engineering

Second-degree interaction features were generated using PolynomialFeatures with interaction_only=True and include_bias=False. This allowed the model to test whether combinations of numerical student characteristics provided additional predictive information.

4.4 Hyperparameter Tuning

GridSearchCV with 5-fold cross-validation and F1 scoring was used. Logistic Regression was tested over different C values and solvers. Random Forest was tuned using n_estimators, max_depth, and min_samples_split. SVM was tuned using C, kernel, and gamma. KNN was tuned using n_neighbors, weights, and distance metric.

5. Experimental Results

The following results were obtained on the held-out test set. The models are ordered by F1 score, which was used as the primary comparison metric because it balances precision and recall.

Rank	Model	Accuracy	Precision	Recall	F1 Score
1	**Week 3 Baseline Logistic Regression**	**80.85%**	**76.95%**	**77.59%**	**77.27%**
2	Default SVM	80.65%	77.83%	75.33%	76.56%
3	Feature Engineered Logistic Regression	80.25%	77.14%	75.21%	76.16%
4	Tuned Logistic Regression	80.20%	77.05%	75.21%	76.12%
5	Tuned SVM	80.15%	77.15%	74.85%	75.98%
6	Tuned Random Forest	79.95%	77.31%	73.90%	75.56%
7	Tuned KNN	79.35%	75.24%	75.69%	75.46%
8	Default Random Forest	79.45%	76.88%	72.94%	74.86%
9	Default KNN	77.60%	74.29%	71.28%	72.75%

6. Analysis and Discussion

6.1 Feature Engineering Result

The feature-engineered Logistic Regression model achieved 80.25% accuracy and a 76.16% F1 score, compared with 80.85% accuracy and 77.27% F1 for the original baseline. Therefore, the interaction features did not provide a net improvement on the held-out test data. Precision increased slightly, but recall and F1 decreased.

6.2 Logistic Regression Tuning

Hyperparameter tuning produced an accuracy of 80.20% and an F1 score of 76.12%. This was slightly below the baseline. The result indicates that the original Logistic Regression configuration was already well suited to the available feature representation.

6.3 Random Forest

Default Random Forest achieved 79.45% accuracy and 74.86% F1. After tuning, performance increased to 79.95% accuracy and 75.56% F1. Thus, hyperparameter optimization improved Random Forest, but it did not surpass Logistic Regression.

6.4 SVM

Default SVM achieved 80.65% accuracy and 76.56% F1, making it the second-best overall model by F1. Its precision of 77.83% was the highest among the evaluated models. However, tuning reduced its test F1 to

75.98%.

6.5 KNN

Default KNN produced the lowest performance, with 77.60% accuracy and 72.75% F1. Tuning improved its F1 to 75.46% and recall to 75.69%, showing that parameter selection was useful, although the model still did not match the stronger approaches.

7. Final Model Selection

The Week 3 Baseline Logistic Regression model is selected as the final model because it achieved the highest test accuracy (80.85%), highest recall (77.59%), and highest F1 score (77.27%) among all experiments. Although other models achieved slightly higher precision, none produced a better overall balance of precision and recall.

Final recommendation: retain the original Logistic Regression model for the current Student Placement Prediction system. The Week 4 experiments demonstrate that additional complexity should not be introduced unless it produces measurable improvement under robust validation.

8. Limitations and Future Improvements

The experiments were evaluated using one held-out test split. Future work can improve the reliability of the conclusions through repeated or stratified cross-validation, broader hyperparameter searches, probability calibration, threshold optimization, class-imbalance analysis, and additional domain-informed features.

Potential future algorithms include Gradient Boosting, XGBoost or other tree-boosting methods where appropriate. Feature selection could also be explored to remove redundant variables and simplify the final model.

9. Conclusion

Week 4 followed an iterative machine-learning workflow by extending the Week 3 baseline, engineering interaction features, tuning hyperparameters, and comparing multiple algorithms. The results show that more complex models do not automatically provide better performance. For this Student Placement Prediction dataset, the original Logistic Regression model remained the most effective overall approach, achieving 80.85% accuracy and a 77.27% F1 score.

The experiments nevertheless provided valuable evidence about the behavior of each algorithm and demonstrated a systematic model-development process suitable for real-world machine-learning work.

End of Report